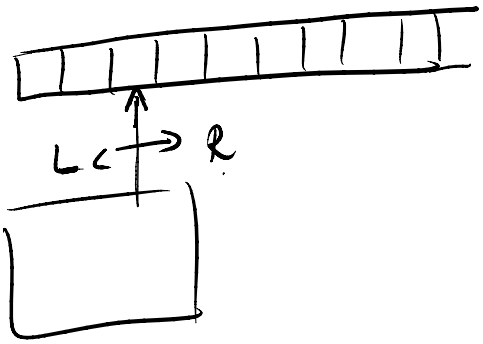
Seminar10

luni, 10 ianuarie 2022 12:08

Masina Turing



$M = (K, \Sigma, \delta, s)$

Operatii ale MT.

$a, b, c, \$$

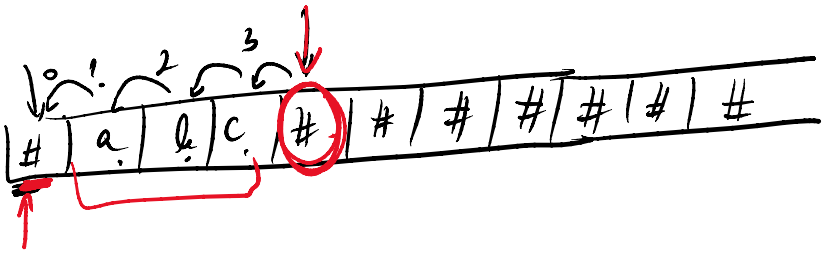
L.

R.

$\forall a \in \Sigma$, a.

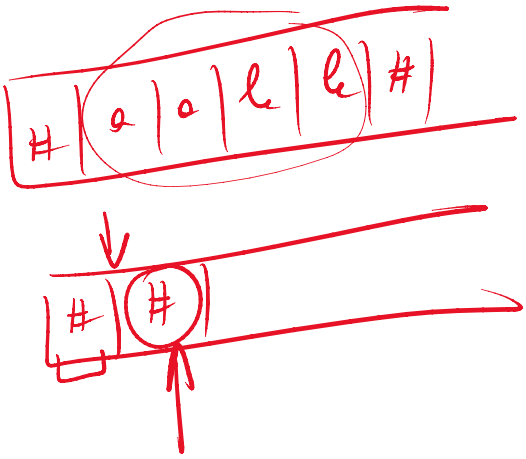
b.

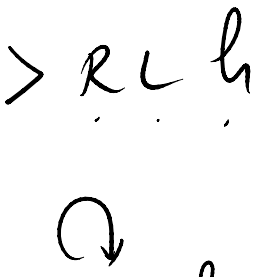




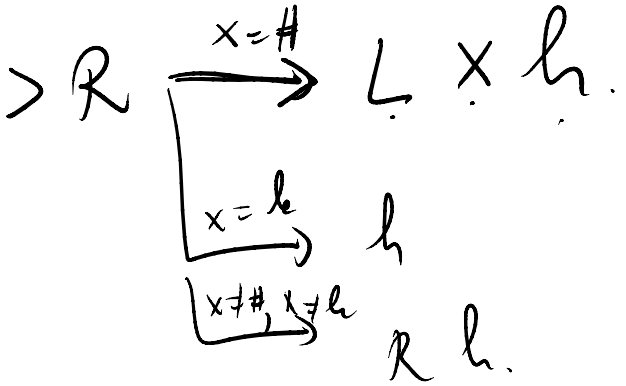
a b c

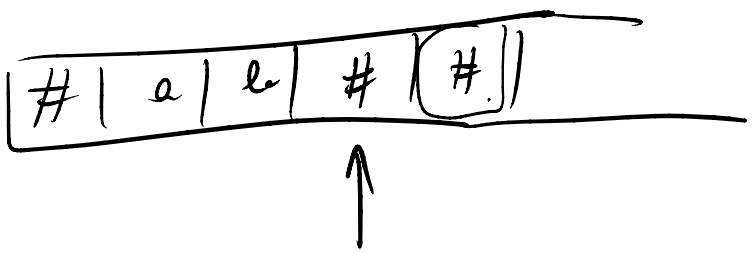
MT → copiază











M_1

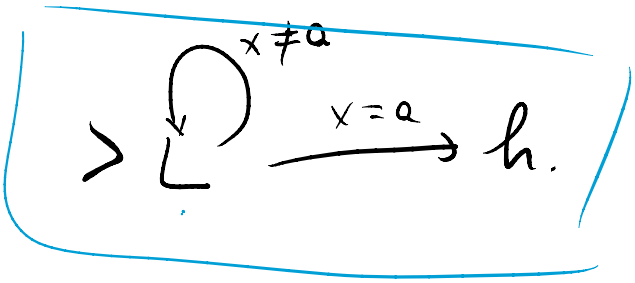
M_2

$|> M_1 \; R \; M_2 \; h$

L_a , R_a , $L_{\bar{a}}$, $R_{\bar{a}}$

L_a stânga până la simbolul a.

$\# \boxed{a} \; l \; c \; l \; \#$

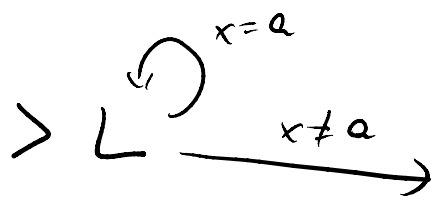


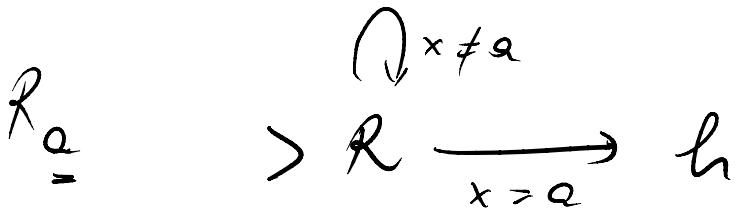
L_a.

L_1

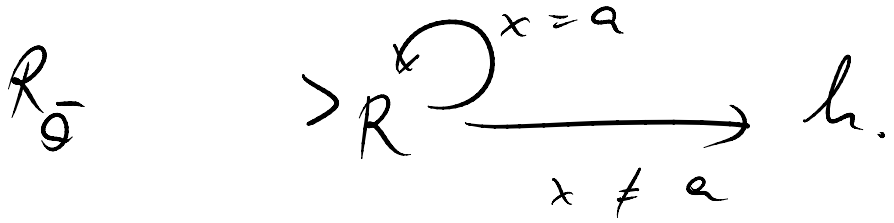
$L_{\bar{a}}$ stânga cât timp găsește a

$\# \boxed{l} \; a \; a \; \#$



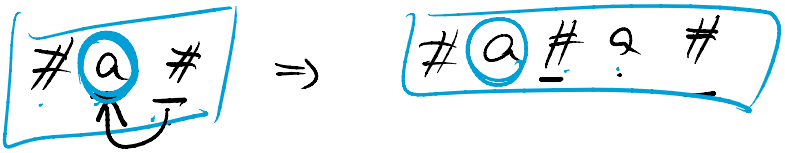


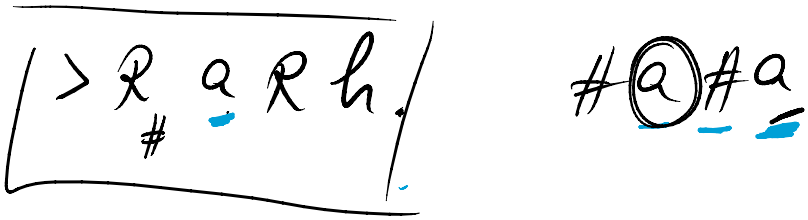
R_a



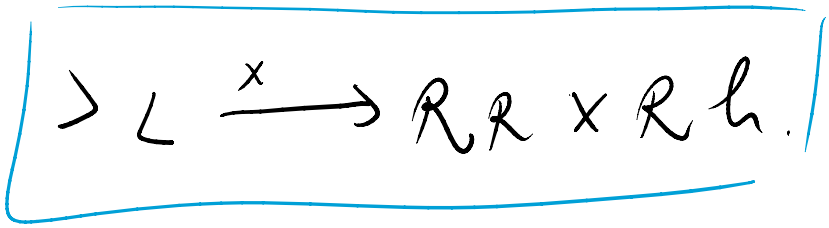
---

1. Scrieți o MT care copiază simbolul de pe bandă:





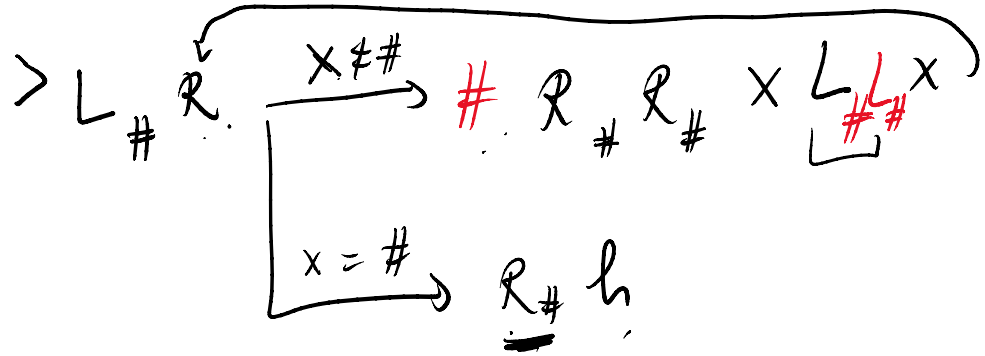
a # a

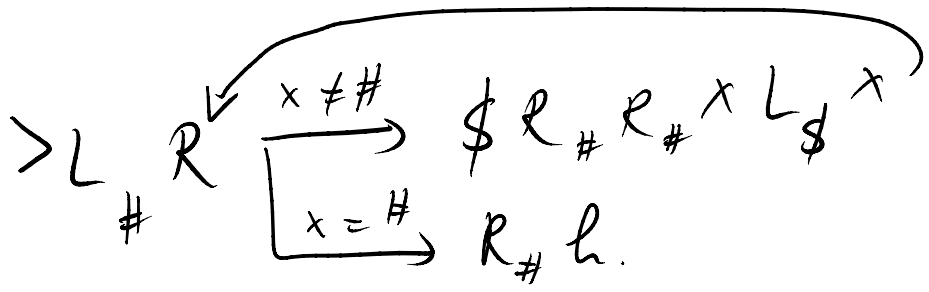


2. MT care copiază un cuvânt

x # ⇒ # x # x

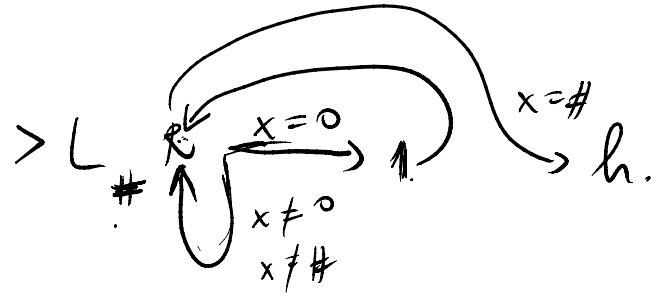
a b c # a b c





4. MT care înlocuiește orice număr 0 cu 1.

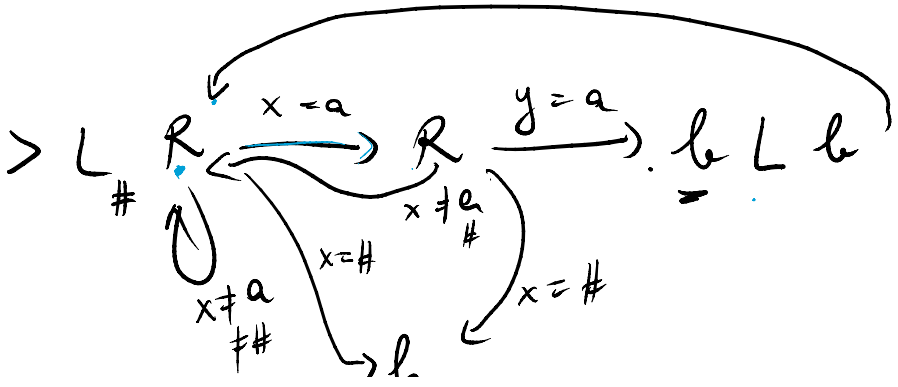
1.1 1 1 1.1 # ⇒ # 1 1 1 1 1 1

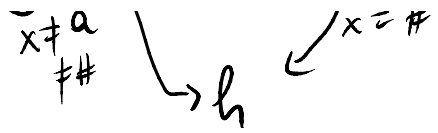


6. MT care înlocuiește aa cu bb

aa b a b aaa

⇒ # bb b a b bb a #

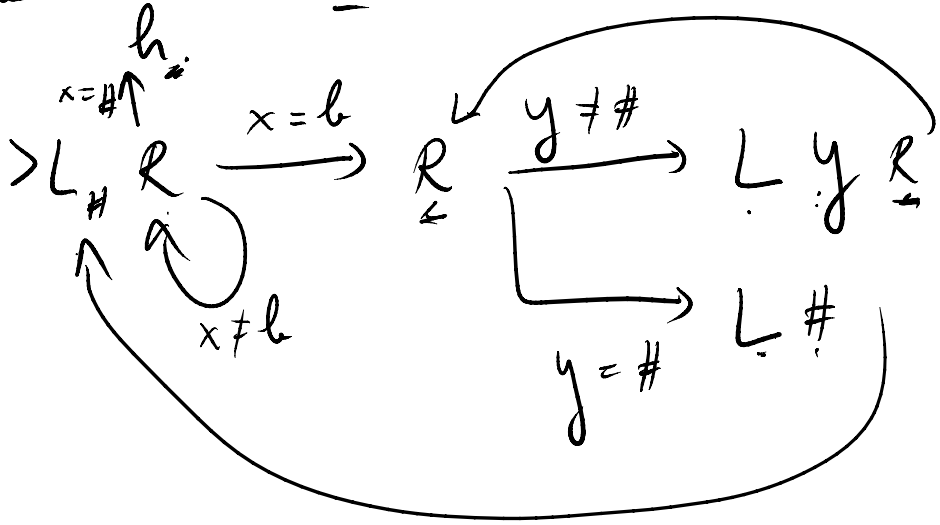




b b b a #

6. O MT care șterge toate b-urile din cuvânt.

a b c c a b # ⇒ # a c c a



a c # #

7. O MT care incrementează val. în binar

1 0 # ⇒ # 1 1

1 1 # → # 1 0 0

0, 1, 10, 11, 100, 101, ...